

The only entity with redundancy is SubwayTrain:

SubwayTrain(VehicleID: number(10), TrainModel: char(40), NumberOfCars: number(2), TrainSetNumber: char(20))

Where the FDs are:

1. VehicleID -> TrainModel, NumberOfCars, TrainSetNumber, Manufacturer
2. TrainSetNumber -> VehicleID, TrainModel, NumberOfCars, Manufacturer
3. TrainModel->NumberOfCars

Only 3 violates BCNF

After normalization we get

SubwayTrain(VehicleID: number(10), TrainModel: char(40), TrainSetNumber: char(20))

And

TrainModel(TrainModel: char(40), NumberOfCars: number(2)) where TrainModel is the primary key

Both SubwayTrain and TrainModel follows BCNF